1.(i) Implement the Candidate-Elimination Algorithm using a given training dataset.
(ii) Determine and display the set of all hypotheses consistent with the training examples.

Sky	**Air Temp**	**Humidity**	**Wind**	**Water**	**Forecast**	**Enjoy Sport**
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

2.(i) Build an Artificial Neural Network using the Backpropagation Algorithm.

(ii) Test and evaluate the network using an appropriate dataset.

3.(i) Implement the k-Nearest Neighbour algorithm for classifying the Pulmonary dataset.

(ii)Display and analyse both correct and incorrect predictions.

Patient ID	**Age**	**Cough Severity (1-10)**	**Oxygen Level (%)**	**Disease**
P1	25	2	98	Healthy
P2	38	5	94	Mild
P3	45	6	92	Mild
P4	58	8	88	Severe
P5	65	9	84	Severe

4.(i) Apply the k-Means clustering algorithm for the pulmonary diseases data set

(ii) Evaluate and comment on the quality of the clustering results.

Patient ID	Age	Respiratory Rate (breaths/min)	Oxygen Saturation (%)
P1	25	16	98
P2	30	18	97
P3	40	22	94
P4	45	24	92
P5	55	28	88

5.(i)Perform Principal Component Analysis on a given dataset.

(ii) Interpret and discuss the PCA results.

Mushroom ID	Cap Diameter (cm)	Stem Height (cm)	Stem Width (cm)	Weight (g)
M1	4.5	6.0	1.2	20
M2	5.0	6.5	1.3	25
M3	6.2	7.0	1.5	32
M4	7.0	8.0	1.8	40
M5	8.0	9.0	2.0	48

6(i) Implement the Naïve Bayesian classifier using a mushroom dataset

(ii) Test the classifier on sample datasets and compute its accuracy.

Cap Colour	**Odor**	**Gill Size**	**Habitat**	**Edible**
Brown	Pleasant	Broad	Forest	Yes
White	Foul	Narrow	Grassland	No
Red	Pleasant	Broad	Forest	Yes
Brown	Foul	Narrow	Urban	No
White	Pleasant	Broad	Grassland	Yes

7(i) Build an Artificial Neural Network using the Backpropagation Algorithm.

(ii) Test and evaluate the network using an appropriate dataset.

Patient ID	**Age**	**Cough Severity (1-10)**	**Oxygen Level (%)**	**Disease**
P1	25	2	98	Healthy
P2	35	4	95	Healthy
P3	45	6	92	Mild
P4	55	8	88	Severe
P5	65	9	84	Severe

8(i) Implement the Candidate-Elimination Algorithm using the Online Course Enrolment dataset.

(ii) Find the specific hypothesis (S) and general hypothesis (G) consistent with the training examples.

Programming Knowledge	**Internet Access**	**Interest Level**	**Time Available**	**En roll Course**
Yes	Good	High	Yes	Yes
Yes	Good	Medium	Yes	Yes
No	Good	High	Yes	No
Yes	Poor	High	No	No
Yes	Good	High	No	Yes

9(i)Implement the Naïve Bayesian classifier using a Industrial Manufacturing dataset

(ii)Test the classifier on sample datasets and compute its accuracy.

Machine Type	**Temperature**	**Vibration**	**Shift**	**Product Quality**
CNC	High	High	Night	Defective
CNC	Normal	Low	Day	Good
Lathe	Normal	Low	Day	Good
Milling	High	High	Night	Defective
Lathe	Normal	High	Day	Good

10(i)Perform Principal Component Analysis on your own dataset.

(ii) Interpret and discuss the PCA results.

11(i) Apply the k-Means clustering algorithm for the pulmonary diseases of appropriate dataset.

(ii) Evaluate and comment on the quality of the clustering results.

12(i) Implement the k-Nearest Neighbour algorithm for classifying the mushroom dataset.

(ii) Display and analyse both correct and incorrect predictions.

Mushroom ID	Cap Diameter (cm)	Stem Height (cm)	Weight (g)	Type
M1	4.5	6.0	20	Edible
M2	5.0	6.5	25	Edible
M3	6.2	7.0	32	Poisonous
M4	7.0	8.0	40	Poisonous
M5	4.8	6.2	22	Edible

13 (i)Build an Artificial Neural Network using the Backpropagation Algorithm using Backpropagation Algorithm.

(ii) Test and evaluate the network using the given dataset.

Temperature (°C)	Pressure (bar)	Vibration Level	Defect
80	30	Low	No
95	45	High	Yes
85	35	Low	No
100	50	High	Yes
90	40	Medium	No

14**(i):** Implement the Candidate-Elimination Algorithm using the Student Scholarship dataset.

(ii) Find the specific and general hypotheses that are consistent with the training examples.

CGPA	Attendance	Family Income	Extra Activities	Scholarship
High	Good	Low	Yes	Yes
High	Good	Medium	Yes	Yes
Medium	Good	Low	No	Yes
Low	Average	High	No	No
Medium	Average	Medium	No	No

15(i) Apply the k-Means clustering algorithm to the Cyber Attack Detection dataset.

(ii) Group the devices into clusters and analyse whether they represent Normal, Suspicious, or Attack behaviour.

Failed Login Attempts	Network Traffic (MB)	Suspicious Requests
2	150	5
15	850	40
3	180	8
20	950	50
8	400	20

16(i) Implement the k-Nearest Neighbour algorithm for classifying the Iris dataset.
(ii) Display and analyse both correct and incorrect predictions.

Sepal Length (cm)	Sepal Width (cm)	Petal Length (cm)	Petal Width (cm)	Species
5.1	3.5	1.4	0.2	Setose

Sepal Length (cm)	Sepal Width (cm)	Petal Length (cm)	Petal Width (cm)	Species
4.9	3.0	1.4	0.2	Setose
6.0	2.9	4.5	1.5	Versicolor
6.5	3.0	5.8	2.2	Virginica
5.7	2.8	4.1	1.3	Versicolor

17(i)Build an Artificial Neural Network using the Backpropagation Algorithm.

(ii)Test and evaluate the network using an appropriate dataset.

18(i) Implement the Candidate-Elimination Algorithm using the Loan Approval dataset.

(ii) Find the specific hypothesis (S) and general hypothesis (G) that are consistent with the training examples.

Income	Employment	Credit Score	Property	Loan Approved
High	Permanent	Good	Yes	Yes
High	Permanent	Average	Yes	Yes
Low	Temporary	Poor	No	No
Medium	Permanent	Good	No	Yes
Low	Permanent	Average	No	No